

Programming Exercise

The goal of this practice is to code a function that generates a sample of the Barabasi-Albert network model with a given network size N and minimum degree m , and to check some of its statistical properties.

Background

The Barabási-Albert model (BA) was the first non-trivial network model designed to explain the origin of the ubiquitous power-law degree distributions observed in real complex networks. Based in the preferential attachment mechanisms, it considers a growing network in which, at each time step, a new node is added, node that is connected to other m nodes already present in the network. The BA model is defined as follows: We start with an initial nucleus of $m + 1$ vertices forming a clique (every one connected to the others). The network grows in time by adding every time step a new vertex, which is connected to m old vertices. The end point of each edge is connected to vertex i in the network with probability

$$\Pi_i = \frac{k_i}{\sum_j k_j}, \quad (1)$$

that is, with probability proportional to the degree of each old vertex. This process is iterated until the network has the predefined size N .

The BA model can be analytically solved by various means. A simple mean-field approximation in the continuous degree limit yields a degree distribution

$$P_{MF}(k) = \frac{2m^2}{k^3}. \quad (2)$$

A more refined master equation approach leads to the more exact form, in the limit of infinite network size

$$P_\infty(k) = \frac{2m(m+1)}{k(k+1)(k+2)}. \quad (3)$$

For small network sizes, the numerical degree distribution presents finite-size corrections with respect to the analytical expression, which take the form

$$P_N(k) = P_\infty(k)w\left(\frac{k}{\sqrt{N}}\right), \quad (4)$$

where $w(x)$ is a universal function, independent of k and N .

The computer project

1. Code a function (subroutine) that generates the BA model for an arbitrary value of N and m . To store the network structure, it is useful to use an adjacency list, that is, a

list of lists that associates to each node a list with its nearest neighbors. Instead of a list of lists, a dictionary can be used, that for each key, corresponding to the index of each node, assigns as value the list of its nearest neighbors. Lists are useful since they can grow in the process of generating the BA network. For the generating process, one needs to select old nodes with probability Eq. (1) to connect the newly added nodes. Sampling from such changing distribution can be done using an auxiliary list to select the vertices to which new edges are attached. Add to this array the number i , every time you add a new connection to vertex i , and m times the number j whenever you add the new vertex j . Any number i will appear in this way k_i times. Selecting uniformly at random an element in this auxiliary array will give you the number i with probability proportional to k_i . Do not forget the initial nucleus of the network. You can use the random number generator provided by Python (the high-quality Mersenne-Twister).

2. For $m = 4$, generate one BA network of large size, $N = 10^6$ if possible or larger, and compute its degree distribution. Make a plot showing that the prediction Eq. (3) is better than the approximation Eq. (4, specially for small values of the degree k .
3. For $m = 4$, compute the average degree distribution $P_N(k)$ for BA networks with size $N = 50, 100, 200$ and 500 . Perform averages over a number of $M = 10000$ different networks. For the averages, make a histogram accumulated the degrees of all the networks generates and apply normalization at the end. Check the finite size corrections in Eq. (4) by plotting, for the different values of N the $P_{est}(k)/P_\infty(k)$ as a function of k , where $P_{est}(k)$ is the average degree distribution that you have estimated numerically.
4. Extra point: For a minimum degree $m = 4$, compute the average clustering coefficient $\langle c \rangle_N$ for BA networks with size $N = 25, 100, 200, 500, 1000, 5000$ and 10000 . Perform averages over $M = 25$ different networks. Plot the average clustering coefficient as a function of the network size.

References